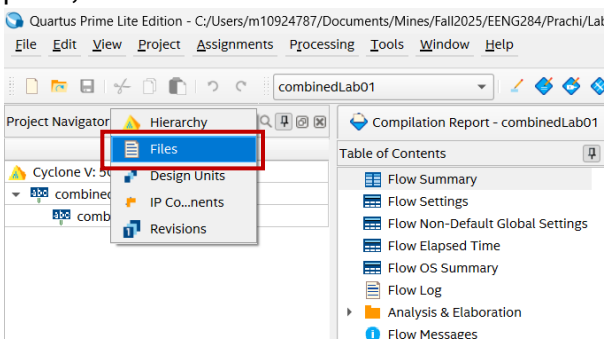
The steps are outlined using Lab02 part: combinedLab01 as an example. The `combinedLab01` Verilog file has three inputs and three outputs that are mapped to slide switches and LEDs.

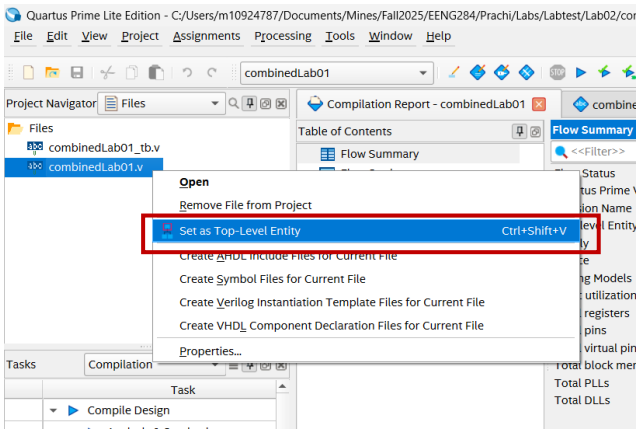
Synthesizing a Verilog Module

It's time to realize the `combinedLab01` Verilog file to FPGA. To do this follow these steps:

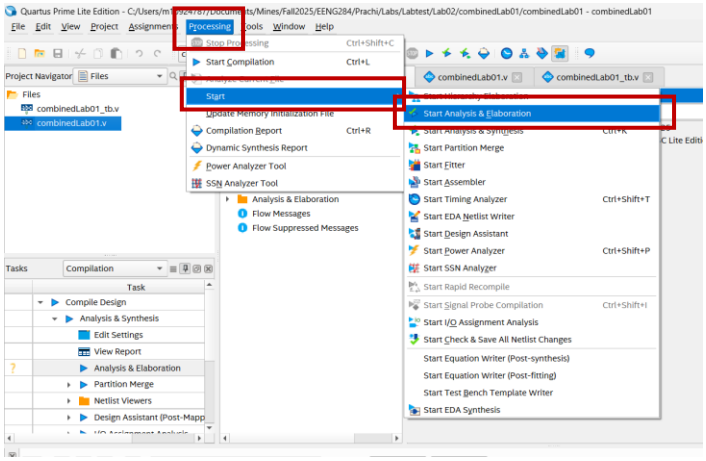
1. In Project Navigator pane, select the File tab.



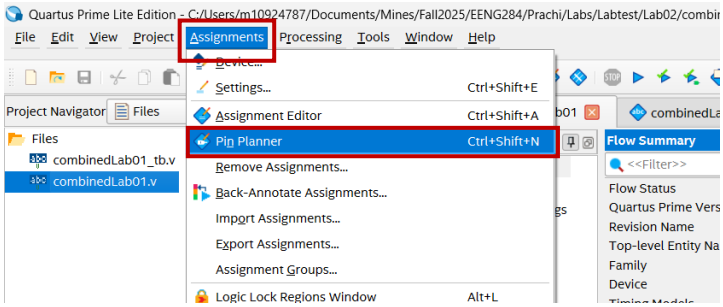
2. Right mouse click `combinedLab01.v` and select Set as Top-Level Entity.



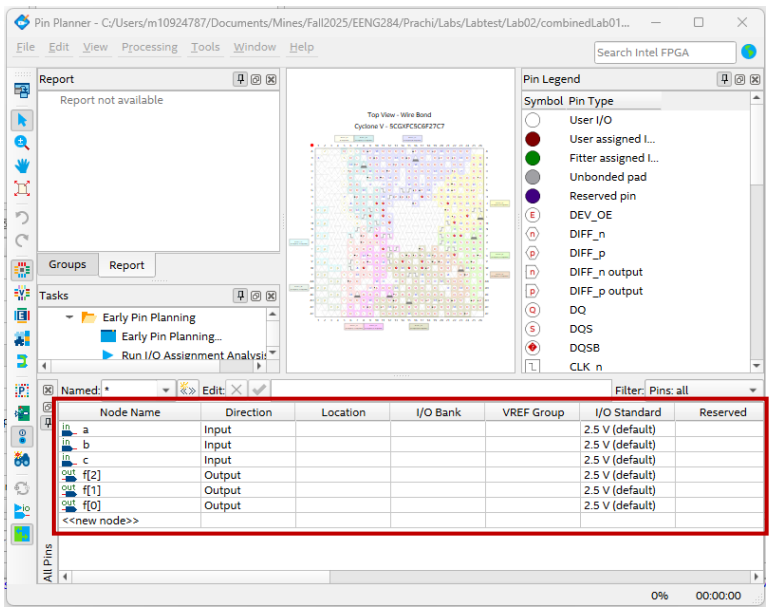
3. Processing -> Start -> Start Analysis and Elaboration



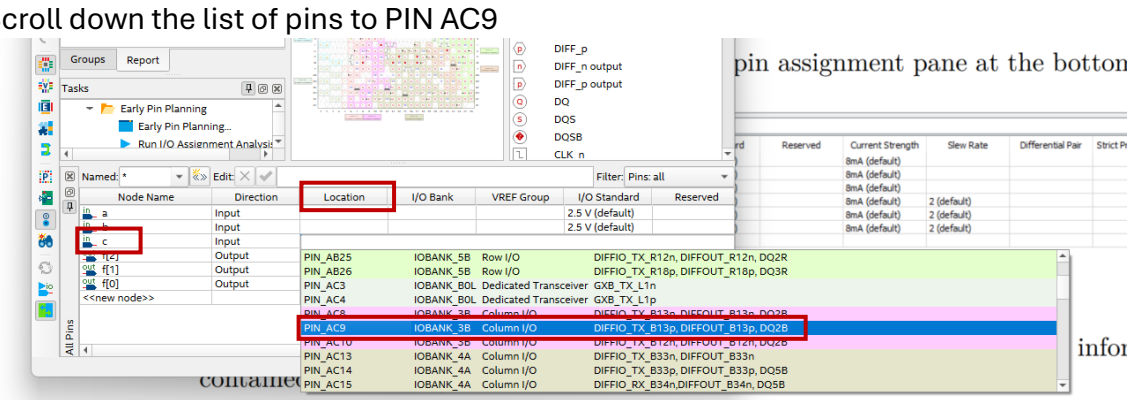
4. Assignments -> Pin Planner



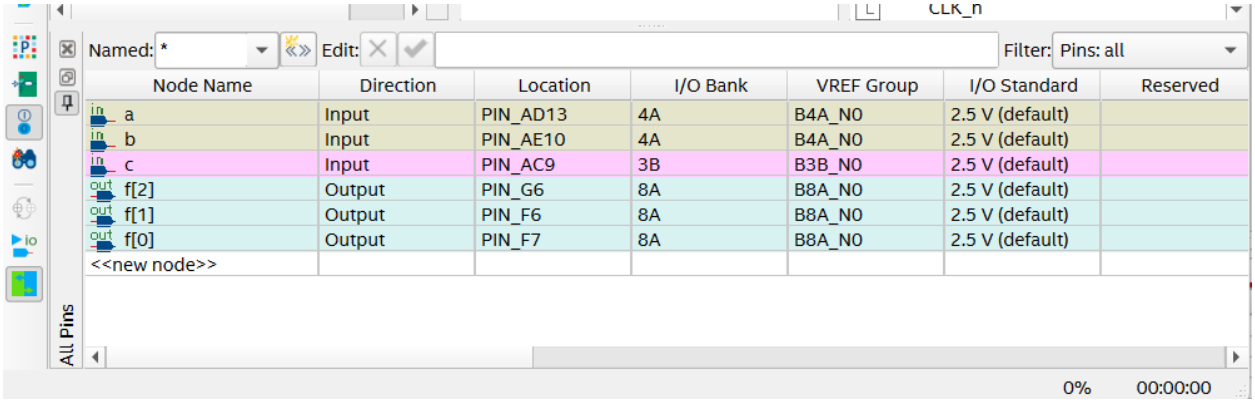
5. In the Pin Planner pop-up you should see the pin assignment pane at the bottom of the window.



6. Double click in the Location cell for row c.
7. Scroll down the list of pins to PIN AC9



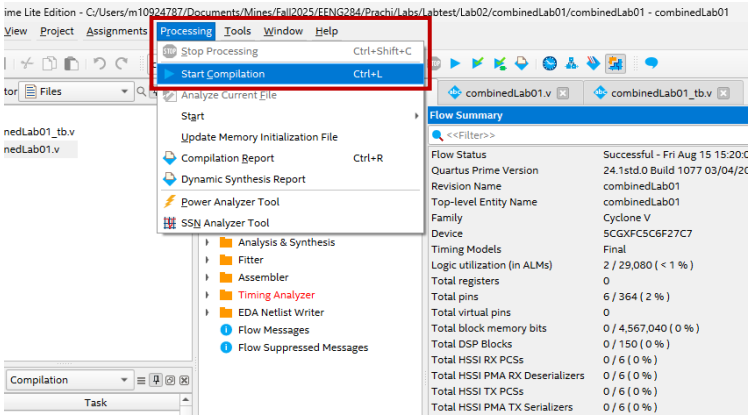
8. Complete the pin assignment for the other 5 inputs and outputs using the information from the C5G User Manual.



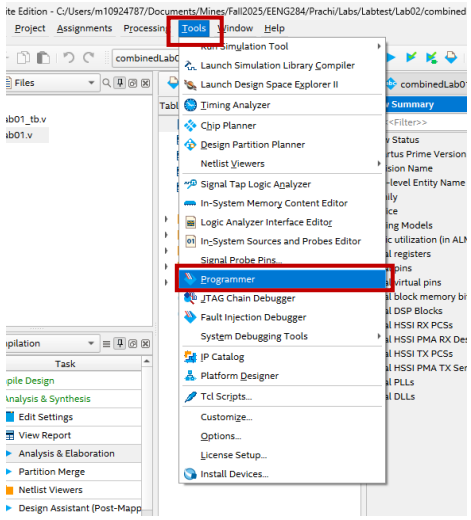
Node Name	Direction	Location	I/O Bank	VREF Group	I/O Standard	Reserved
a	Input	PIN_AD13	4A	B4A_N0	2.5 V (default)	
b	Input	PIN_AE10	4A	B4A_N0	2.5 V (default)	
c	Input	PIN_AC9	3B	B3B_N0	2.5 V (default)	
f[2]	Output	PIN_G6	8A	B8A_N0	2.5 V (default)	
f[1]	Output	PIN_F6	8A	B8A_N0	2.5 V (default)	
f[0]	Output	PIN_F7	8A	B8A_N0	2.5 V (default)	
<<new node>>						

9. Double check your pin assignments.
10. File -> Close. Note closing your file incorporates this assignment into the project.

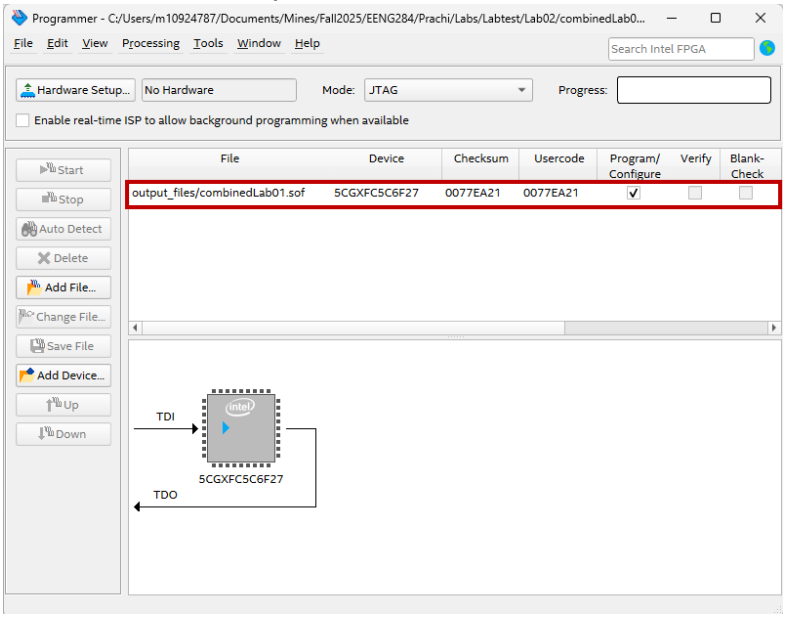
11. Back in the Quartus window, Processing -> Start Compilation



12. Tools -> Programmer

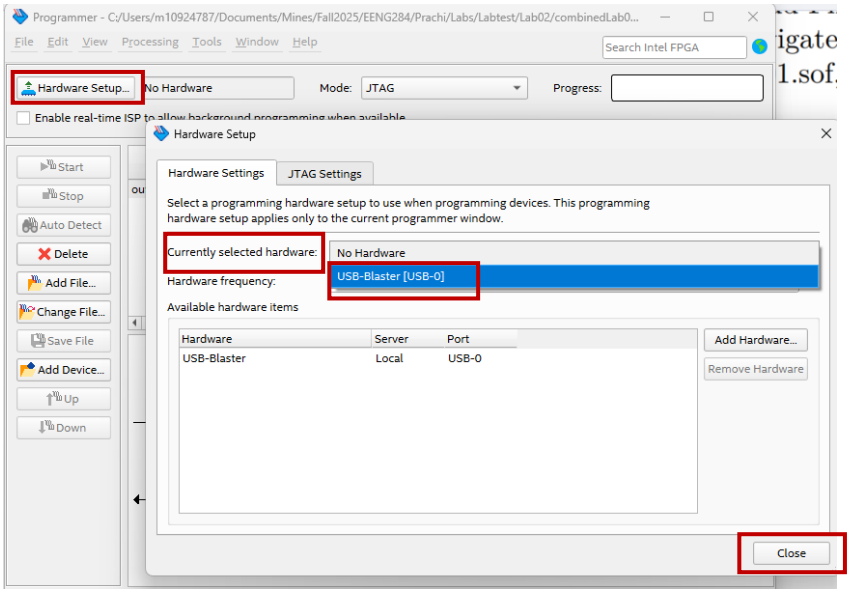


13. In the Programmer pop-up window click Add File. . .
14. In the Select Programming File pop-up, navigate to your project directory, then into the output files folder, the select `combinedLab01.sof`, click Open. You should see something like the following. The .sof file may already show up as added in which case no need to do this step.

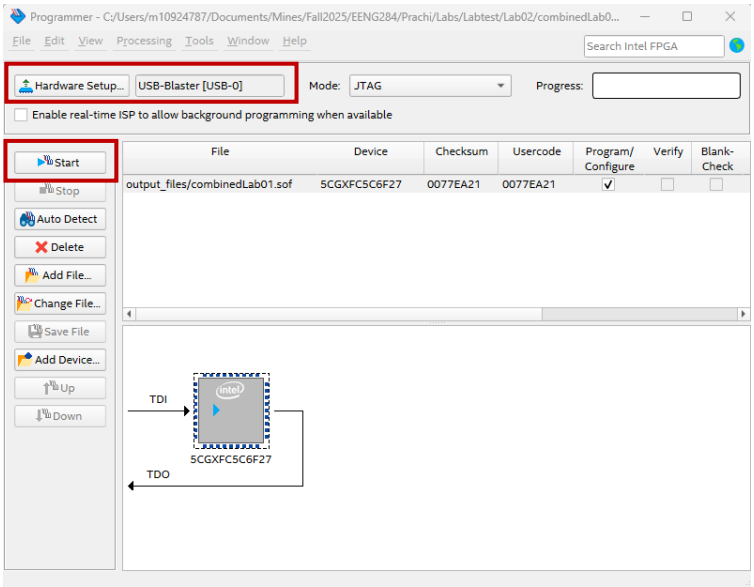


15. Connect the Altera Cyclone V GX FPGA to your computer through the USB Blaster port, connect the power supply, and push the red power-on button. Try not to be annoyed by the infernal blinking LEDs.
16. In the Programmer pop-up (The first time you run synthesis, and don't see the USB Blaster in Hardware, go to end of the document to see additional steps to be followed.)

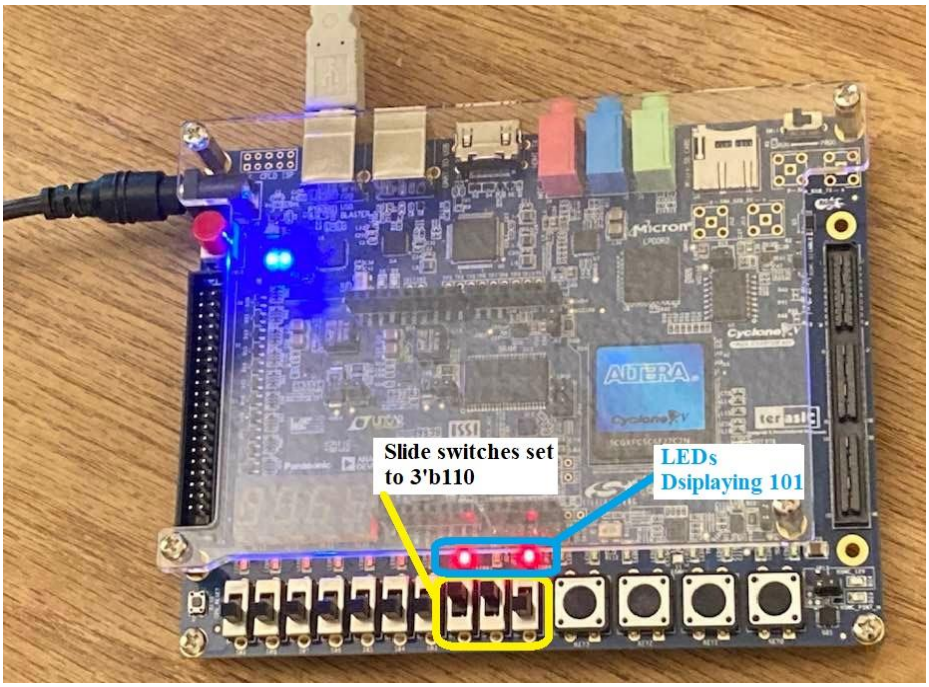
a. Click Hardware Setup. . .
b. In the Hardware Setup select USB-Blaster [USB-0] from the Currently selected hardware pull-down
c. Click Close



17. Back in the Programmer window, the box next to Hardware Setup. . . should reflect your choice. Click Start.



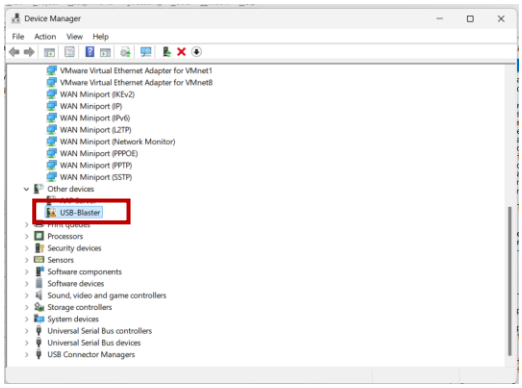
18. The Development board should stop its infernal blinking and run your program. You may notice that the unused LEDs are dimly illuminated.
19. Move the slide switches up and down to verify that the input/output matches the logic function you implemented. Use white silk screen printing on the development board to locate slide switches and LEDs you assigned in your pin-assignment.



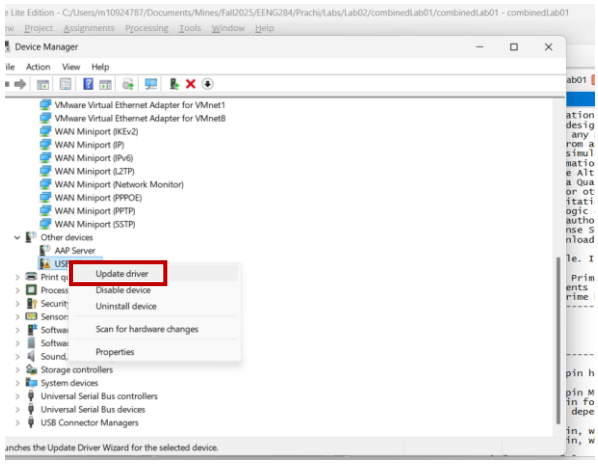
The Development board properly configured and running the `combinedLab01` Verilog file.

If you don't see USB Blaster as an option in the hardware set-up, it could be because the driver is not installed. Follow these steps (you should only have to do these the first time you synthesize code). Make sure the board is connected to your laptop through USB Blaster port.

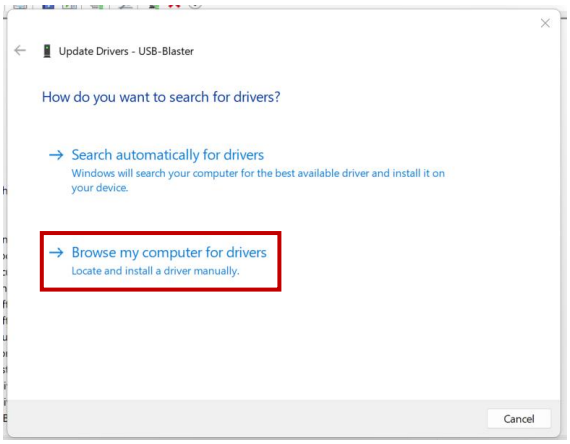
1. Open Device Manager on your laptop. Next to USB Blaster, you will see a ⚠ symbol.



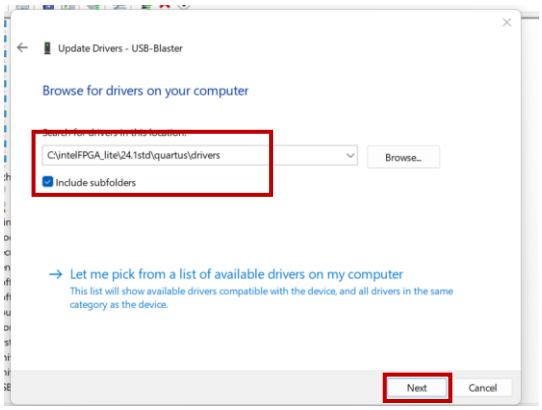
2. Right click on USB-Blaster and Update Driver

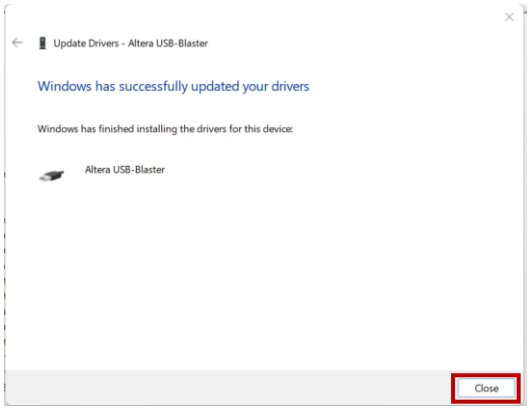


3. Browse my computer for drivers

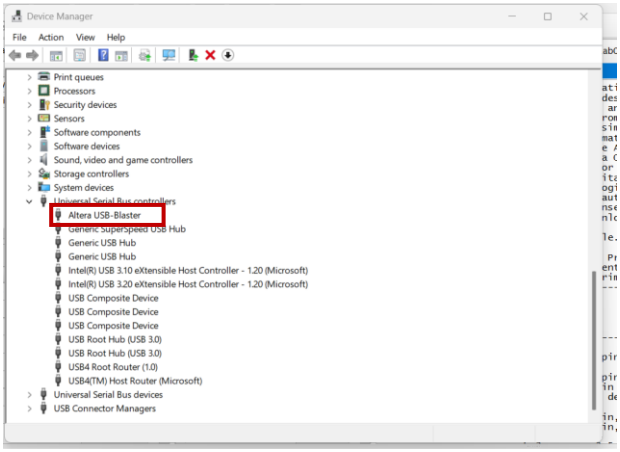


4. Navigate to C:\intelFPGA_lite\24.1std\quartus\drivers and check 'Include subfolders' option.





5. You should now see the Altera USB-Blaster in Universal Serial Bus controllers



Go back to step 16, you should see the USB-Blaster in the hardware set up drop down